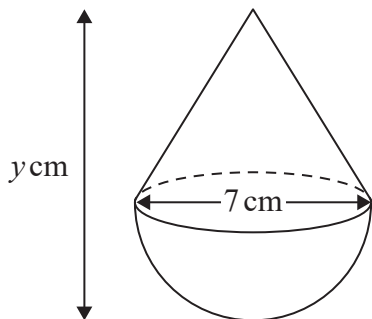
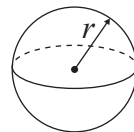


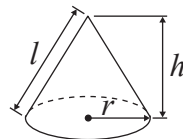
17 A solid cone is joined to a solid hemisphere to make the solid **T** shown below.



$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$



$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$



The diameter of the base of the cone is 7 cm.

The diameter of the hemisphere is 7 cm.

The total volume of **T** is $120\pi \text{ cm}^3$

The total height of **T** is $y \text{ cm}$.

(a) Calculate the value of y .

Give your answer correct to 3 significant figures.

$$y =$$

(4)

The diameter of the base of the cone and the diameter of the hemisphere are both increased by the same amount.

Assuming the total volume of **T** does not change,

(b) explain the effect this would have on your answer to part (a).

(1)

(Total for Question 17 is 5 marks)